CS101A DATA STRUCTURE(H)

QUIZ 5

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Start Time: 20:40 PM **End Time:** 21:00 PM

Q. 1 TRUE / FALSE (18 pts)

1. If DFS from s visits all vertices, BFS from s may not visit all vertices. ○ **T** ● **F**

2. If all edge weights in a connected graph are distinct, the MST may not unique. ● **T** ● **F**

3. If a graph has two different minimum spanning trees, then they must have the same total weight. ● **T** ○ **F**

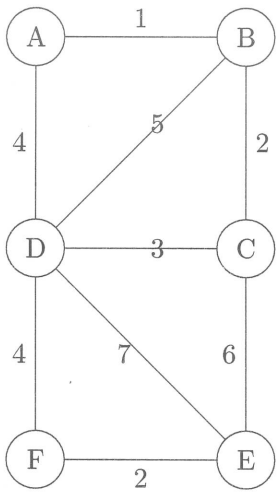
4. A directed graph has a topological ordering if and only if it has no directed cycle. ○ **T** ● **F**

5. If a directed graph has a topological ordering, then removing any single edge keeps the graph having a topological ordering. ● **T** ○ **F**

6. In a DAG, a vertex v can appear first in some topological ordering if and only if v has no incoming edges. ● **T** ○ **F**

Q. 2 MINIMUM SPANNING TREE — KRUSKAL & PRIM (24 pts)

Consider the connected undirected weighted graph G shown below. When multiple edges have the same weight, process them in lexicographic order of endpoints.



(a) (Kruskal simulation) (10 pts)

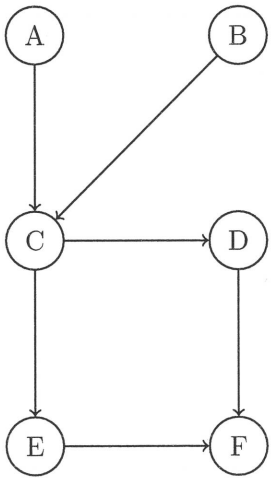
Fill in the table. "Selected?" is **Y** if the edge is added to the MST, otherwise **N**.

Step	Edge considered	Selected? (Y/N)	MST edges so far (in order)
1	A-B	Y	~~{A,B}~~ (A,B)
2	B-C	Y	~~{A,B,C}~~ (A,B)(B,C)
3	E-F	Y	~~{A,B,C,E,F}~~ (A,B)(B,C)(E,F)
4	C-D	Y	~~{A,B,C,E,F,D}~~ {A,B,C,D}{E,F}
5	A-D	N	~~{A,B,C,E,F,D}~~ {A (A,B)(B,C)(E,F)(C,D)
6	D-F	Y	~~{A,B,C,E,F,D}~~ (A,B)(B,C)(E,F)(C,D)(D,F)
7			
8			
9			

(b) (Prim simulation) (10 pts)
Starting from A, fill in the Prim table. "New vertex" is the vertex that gets newly added to the tree by the selected edge.

Iteration	Selected edge	New vertex	Total weight so far
1	A - B	B	1
2	B - C	C	3
3	C - D	D	6
4	D - F	F	10
5	F - E	E	12

Q. 3 TOPOLOGICAL SORT (12 pts)
Consider the directed graph G shown below.



Whenever there are multiple vertices with no incoming edges, we remove the **alphabetically smallest** one.

(a) (8 pts) Fill in the table of the algorithm execution.

Step	Available (no-incoming) vertices	Chosen vertex	Output so far
1	A, B	A	A
2	B	B	A B
3	C	C	A B C
4	D, E	D	A B C D
5	E	E	A B C DE
6	F	F	A B C DE F

(b) (4 pts) Under the same priority rule, the final topological ordering is:

A B C D E F